

VIVA Oral

Give it your lecture notes. Close them. Get examined out loud.

It checks each answer against your own pages, reads you the line you got wrong, and names the page.

You can recognise the answer on the page. Can you say it?

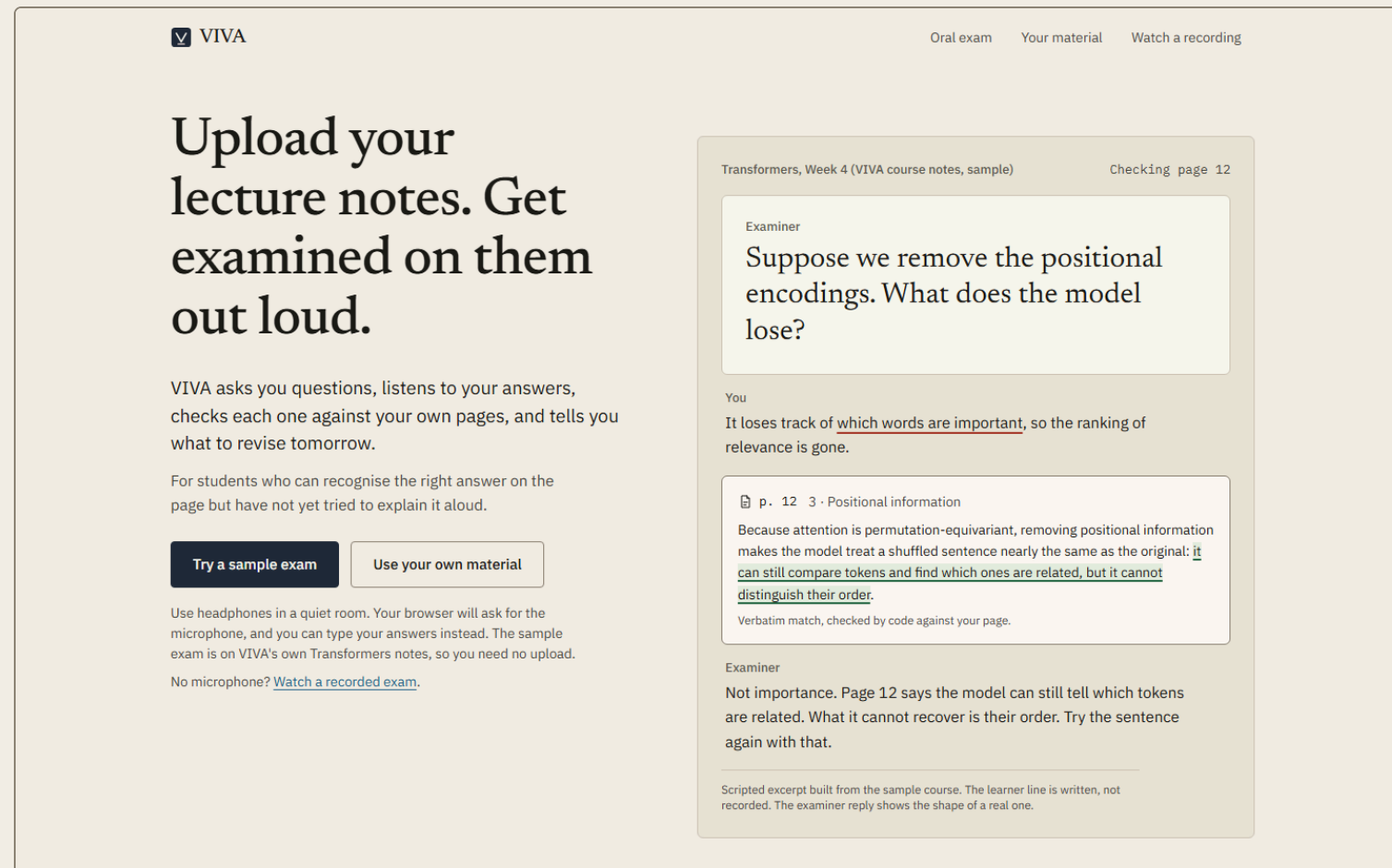
WHO IT HELPS

A student revising from their own text notes before an oral exam, a viva or a seminar, who has read the pages several times and never tried to explain them aloud.

Rereading and highlighting test recognition. An oral exam asks you to produce the explanation with the page closed.

Alone, nobody asks the follow-up. Nobody tells you which page you got wrong, so a confident wrong answer stays wrong until the real exam.

Tools that answer from a model's memory can agree with you when your notes disagree. VIVA only ever quotes your notes, and code checks the quote is really there.



One screen, one job

The front page starts a spoken exam in one click. The panel on the right shows the shape of a session: the question, the learner's wrong phrase marked in red pen, the quoted passage with its page, and the examiner's correction.

WHAT IT IS

Real screenshot, front page, 1440 px, 29 September 2026.

WHAT IS SCRIPTED

The excerpt is built from the sample course. The learner line is written, not recorded.

SAMPLE COURSE

VIVA's own Transformers notes, written for the demo.

The examiner asks. You claim something. A tool checks it before the examiner says a word.

01

Ask

One question at a time, spoken, in the examiner's voice.

02

Answer

You talk. The service decides when your turn ends.

03

Check

The examiner calls `verify_claim` on your exact words.

04

Quote

Only a contradicted verdict allows a correction, with the line and the page.

05

Restate

You say the fact in your own words. You can interrupt at any point.

LEARNER, SYNTHETIC VOICE

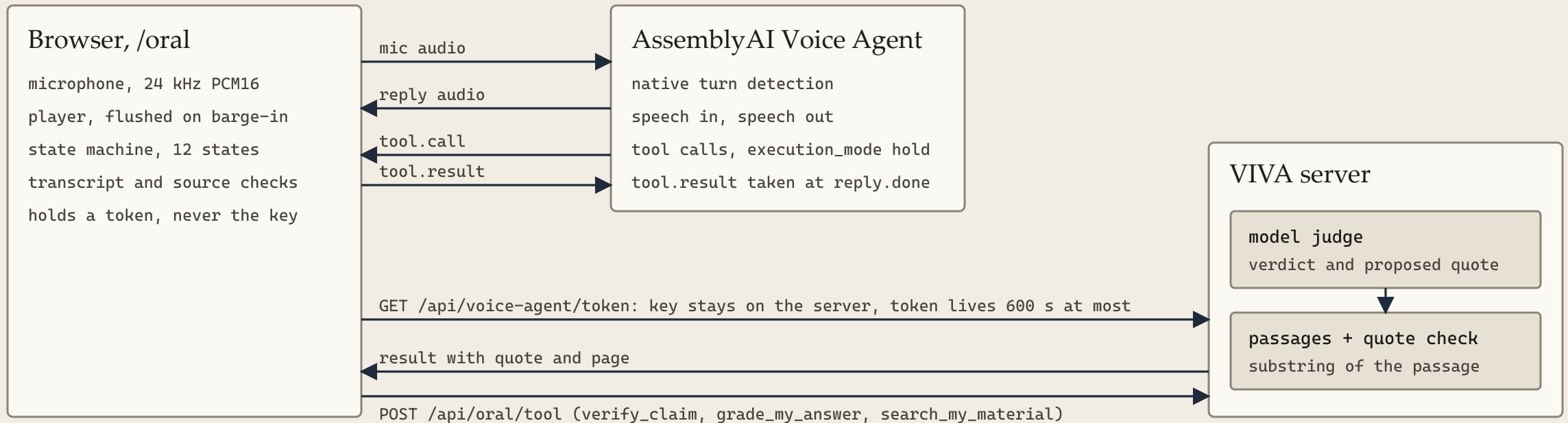
I think it runs a single head over the input.

```
verify_claim(claim, concept: "multi-head attention")  
-> contradicted, page 15, 415 ms judge time
```

EXAMINER, LIVE ASSEMBLYAI SESSION

The material says, Multi-head attention runs the query-key-value computation h times in parallel, each in a smaller subspace, on page fifteen. Please restate that in your own words.

The Voice Agent API runs the conversation.
VIVA supplies the token, the tools and the check.



Code: `src/lib/oral/socket.ts, tools.ts, machine.ts; src/app/api/voice-agent/token; src/app/api/oral/tool`

Reconnect: the service refused `session.resume live`, so VIVA opens a new session with the recent turns

A quotation is allowed only if it is in your notes, word for word.

The model proposes a verdict and a quote. Code checks every piece of the quote is an exact substring of the passage it named, in order. If not, the verdict becomes "not in the material".

yes Real line from page 15, named passage matches. The examiner may quote it and say the page.

no One word of the quote changed, or the quote is from a different passage. Downgraded. The examiner says the material does not settle it.

no The quote is the learner's own words repeated back. Downgraded.

What it prevents

An invented citation, an altered quote, or a page number that does not hold the line. Tested with a mutation that changes one word of a real quote (tests/oral-verify-claim-set.test.ts).

What it does not prevent

A wrong verdict with a real quote. The judge model can be wrong; the substring check cannot see that. On a labelled set of 54 claims the judge made 0 false confirmations, and abstained on 3 claims the labels settled. It is a small set written in this repository.

Live run, 2026-09-29, judge model openai/gpt-oss-120b.

MEASURE	VALUE	N	DATE
Session ready	444 ms	3	2026-09-29
First examiner audio	958 ms	3	2026-09-29
Tool round trip, call to result	2470 ms	3	2026-09-29
Interruption detection: first word to service reports speech	1286 ms	3	2026-09-29
Claim check: labelled claims, false confirmations	54, 0	54	2026-09-29
Claim check: contradicted recall	0.889	54	2026-09-29
Judge time, median and p95	590 and 2067 ms	53	2026-09-29
Front page JS, gzip, throttled mobile profile	139.5 KB	5	2026-09-29
/oral JS, gzip, same profile	178.4 KB	5	2026-09-29

Live rows: real AssemblyAI service, synthetic learner voice (Windows speech synthesis), Node client running the browser's socket code, local server. Not a human on a microphone.

Verified live

- Handshake and first audio, 3 of 3
- Tool call, spoken quote and page, 3 of 3
- Barge-in: 0 stale chunks handed to playback, 6 of 6
- Reconnect as a new session, 2 of 2
- Production: token mint, session config, scripted false claim returns contradicted, page 15

Unit-tested only

- Pending tool result discarded on interruption (live probe failed 3 of 3)
- Twelve machine states, failure screens
- Debrief builder and plan

Not done

- Browser with a human microphone, this build

What it does not do, said before you find it.

- 01 The judge can be wrong. The quote check proves the words are in your notes, not that the verdict is right.
- 02 54 labelled claims over two courses is a small set, written here, with no recorded split between tuning and testing.
- 03 The service reports your interruption 1.3 to 1.5 s after your first word, and VIVA clears the examiner's queued audio when it does. Time to silence in a browser was not measured.
- 04 Code picks your next concept and question kind from your last verdict and your weakest concept. The examiner's wording is the model's: it named that concept in 2 of 3 live runs.
- 05 The next oral session reads your stored map and opens on the weakest concept. History survives a restart only on the Postgres store.
- 06 Text notes only. A scan with no text layer gives it nothing to quote.

ALSO TRUE TODAY

- 01 Storage is durable only with a database. Without one it resets on restart. The deployed build was not yet the final one when this deck was made.
- 02 Legal pages are drafts marked for review.

NOT DONE YET

- OPEN The debrief sheet on screen after an exam
- OPEN A live run of the pending-tool discard
- OPEN A full exam in a browser with a human microphone

Press Try a sample exam. No sign-in, no upload.

1. Open the link and press **Try a sample exam**.
2. Use headphones in a quiet room. Allow the microphone, or type your answers.
3. Say something wrong about multi-head attention and listen to what comes back.

Your microphone audio goes to AssemblyAI while an exam runs. The privacy page lists what is stored and where.

LIVE

<https://viva-five-murex.vercel.app>

SOURCE (MIT)

github.com/vaibhav4046/viva

VIDEO

in the lablab submission

EVIDENCE

`docs/evidence, numbers.json`